

Sequences & Series

ANSWER KEY



Arithmetic sequences: finding terms

- Find the 12th term of the arithmetic sequence with $a = 5$, $d = 3$.
 $a_{12} = 5 + 11(3) = 5 + 33 = 38$.
- Find the 20th term of the arithmetic sequence with $a = -4$, $d = 2$.
 $a_{20} = -4 + 19(2) = -4 + 38 = 34$.
- Find the 15th term of the arithmetic sequence with $a = 100$, $d = -6$.
 $a_{15} = 100 + 14(-6) = 100 - 84 = 16$.
- Write the first four terms of the arithmetic sequence with $a = 7$, $d = -2$.
7, 5, 3, 1.

Arithmetic sequences: finding a and d

- The 4th term of an arithmetic sequence is 17 and the 9th term is 42. Find the first term a and common difference d .
 $a + 3d = 17$ and $a + 8d = 42$. Subtracting: $5d = 25 \Rightarrow d = 5$. Then $a = 17 - 15 = 2$.
- An arithmetic sequence has $a_1 = 10$ and $a_6 = 30$. Find the common difference d .
 $a_6 = a_1 + 5d = 10 + 5d = 30 \Rightarrow 5d = 20 \Rightarrow d = 4$.
- An arithmetic sequence has $a_3 = 11$ and $a_7 = 27$. Find the common difference d .
 $a_7 - a_3 = 4d = 16 \Rightarrow d = 4$.

Arithmetic series: the sum formula

- Find the sum of the first 20 terms of the arithmetic sequence with $a = 3$, $d = 4$, using
 $S_n = \frac{n}{2}(2a + (n - 1)d)$.
 $S_{20} = \frac{20}{2}(6 + 19(4)) = 10(6 + 76) = 10(82) = 820$.
- Find the sum of the first 15 terms of the arithmetic sequence with $a = 50$, $d = -3$.
 $S_{15} = \frac{15}{2}(100 + 14(-3)) = 7.5(100 - 42) = 7.5(58) = 435$.
- Find the sum of the arithmetic series $2 + 5 + 8 + \dots + 50$.
Here $a = 2$, $d = 3$; $50 = 2 + (n - 1)3 \Rightarrow n = 17$. $S_{17} = \frac{17}{2}(2 + 50) = 17(26) = 442$.

- 11 Find the sum of the first 10 positive multiples of 6.

These form an arithmetic sequence with $a = 6$, $d = 6$, $n = 10$.

$$S_{10} = \frac{10}{2}(12 + 9(6)) = 5(12 + 54) = 5(66) = 330.$$

Geometric sequences: finding terms

- 12 Find the 6th term of the geometric sequence with $a = 3$, $r = 2$.

$$a_6 = 3(2)^5 = 3(32) = 96.$$

- 13 Find the 5th term of the geometric sequence with $a = 100$, $r = 0.5$.

$$a_5 = 100(0.5)^4 = 100(0.0625) = 6.25.$$

- 14 Find the 4th term of the geometric sequence with $a = 2$, $r = -3$.

$$a_4 = 2(-3)^3 = 2(-27) = -54.$$

- 15 Write the first four terms of the geometric sequence with $a = 5$, $r = 3$.

5, 15, 45, 135.

Geometric sequences: finding a and r

- 16 The 2nd term of a geometric sequence is 6 and the 5th term is 162. Find r and a .

$$ar = 6 \text{ and } ar^4 = 162. \text{ Dividing: } r^3 = 27 \Rightarrow r = 3. \text{ Then } a = 6/3 = 2.$$

- 17 A geometric sequence has $a_1 = 4$ and $a_4 = 108$. Find r .

$$a_4 = a_1 r^3 = 4r^3 = 108 \Rightarrow r^3 = 27 \Rightarrow r = 3.$$

- 18 A geometric sequence has 3rd term 12 and 6th term 96. Find a and r .

$$ar^2 = 12 \text{ and } ar^5 = 96. \text{ Dividing: } r^3 = 8 \Rightarrow r = 2. \text{ Then } a = 12/4 = 3.$$

Geometric series: the finite sum

- 19 Find the sum of the first 6 terms of the geometric series with $a = 3$, $r = 2$, using $S_n = \frac{a(r^n - 1)}{r - 1}$.

$$S_6 = \frac{3(2^6 - 1)}{2 - 1} = 3(63) = 189.$$

- 20 Find the sum of the first 5 terms of the geometric series with $a = 100$, $r = 0.5$.

$$S_5 = \frac{100(1 - 0.5^5)}{1 - 0.5} = \frac{100(0.96875)}{0.5} = 193.75.$$

- 21 Find the sum of the first 4 terms of the geometric series with $a = 2$, $r = -3$.

$$S_4 = \frac{2(1 - (-3)^4)}{1 - (-3)} = \frac{2(1 - 81)}{4} = \frac{-160}{4} = -40.$$

Infinite geometric series

- 22 Find the sum of the infinite geometric series with $a = 4$, $r = \frac{1}{2}$.
$$S = \frac{a}{1-r} = \frac{4}{1-\frac{1}{2}} = 8.$$
- 23 Find the sum of the infinite geometric series with $a = 9$, $r = \frac{1}{3}$.
$$S = \frac{9}{1-\frac{1}{3}} = \frac{9}{\frac{2}{3}} = 13.5.$$
- 24 Find the sum of the infinite geometric series $8 - 4 + 2 - 1 + \dots$.
Here $a = 8$, $r = -\frac{1}{2}$.
$$S = \frac{8}{1-(-\frac{1}{2})} = \frac{8}{1.5} = \frac{16}{3}.$$
- 25 Express the repeating decimal $0.333\dots$ as a fraction using an infinite geometric series with $a = \frac{3}{10}$, $r = \frac{1}{10}$.
$$S = \frac{\frac{3}{10}}{1-\frac{1}{10}} = \frac{3/10}{9/10} = \frac{3}{9} = \frac{1}{3}.$$
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Sigma notation

- 26 Evaluate $\sum_{k=1}^5 (2k + 1)$.
Terms: $3, 5, 7, 9, 11$. Sum = 35.
- 27 Evaluate $\sum_{k=1}^4 3^k$.
 $3 + 9 + 27 + 81 = 120.$
- 28 Use $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ to evaluate $\sum_{k=1}^{10} k$.
 $\frac{10(11)}{2} = 55.$
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Recursive sequences

- 29 A sequence satisfies $a_1 = 2$ and $a_{n+1} = a_n + 5$. Find a_5 .
 $a_1 = 2, a_2 = 7, a_3 = 12, a_4 = 17, a_5 = 22.$
- 30 A sequence satisfies $a_1 = 3$ and $a_{n+1} = 2a_n - 1$. Find a_4 .
 $a_1 = 3, a_2 = 5, a_3 = 9, a_4 = 17.$
- 31 A sequence satisfies $a_1 = 1$, $a_2 = 1$, and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$. Find a_6 .
 $a_3 = 2, a_4 = 3, a_5 = 5, a_6 = 8.$
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Identifying arithmetic and geometric sequences

- 32 State whether the sequence $4, 9, 14, 19, \dots$ is arithmetic or geometric, and give the common difference or ratio.
Arithmetic, with common difference $d = 5$.

- 33** State whether the sequence 2, 6, 18, 54, ... is arithmetic or geometric, and give the common difference or ratio.
Geometric, with common ratio $r = 3$.
- 34** State whether the sequence 100, 90, 80, 70, ... is arithmetic or geometric, and find its 10th term.
Arithmetic, $d = -10$. $a_{10} = 100 + 9(-10) = 100 - 90 = 10$.
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Reading a sequence from its terms

- 35** The bar chart shows the first five terms of a sequence.
- a** Determine whether the sequence is arithmetic or geometric, and find the common ratio or difference. Geometric — each term is double the previous one, so $r = 2$.
 - b** Find the 8th term. $a_8 = 2 \times 2^7 = 2 \times 128 = 256$.
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Word problems: sequences in context

- 36** This question has two parts. An employee's starting monthly salary is 4000 dollars, and it increases by a fixed amount of 150 dollars each month.
- a** Write a formula for the salary a_n in month n (with $a_1 = 4000$). $a_n = 4000 + (n - 1)(150)$.
 - b** Find the salary in month 12. $a_{12} = 4000 + 11(150) = 4000 + 1650 = 5650$ dollars.
- 37** This question has two parts. A bacteria culture starts with 500 cells and triples every hour. Let a_n be the number of cells after n hours, with $a_0 = 500$.
- a** Write a formula for a_n . $a_n = 500 \times 3^n$.
 - b** Find the number of cells after 4 hours. $a_4 = 500 \times 3^4 = 500 \times 81 = 40500$ cells.